Reading Gender Through Digital Methods: A Digital History Project on Female Representation in *The Rose of Versailles*

Introduction

Birnbaum et al. (2017) associate digital medieval studies with Roberto Busa (1913–2011), a Jesuit theologian who specialised in lexical analysis of Thomas Aquinas's works and used a computerised system to manage the scope of the required search (p. 1). Since then, historians have increasingly turned to digital methods (Nygren et al., 2014, p. 1) to analyse large collections of historical sources, as these tools can help scholars identify patterns and relationships that would otherwise be difficult to detect (Cohen & Rosenzweig, 2006). Nevertheless, many historians remain sceptical of digital methods, questioning whether quantitative approaches can adequately capture the complexity of cultural sources (Milligan, 2022), and researchers often fail to properly maintain research data, such as metadata (Elo, 2020, p. 104). Moreover, in digital archives, colonial biases often shape who gets heard and who remains silenced (Risam, 2019, pp. 47–48).

Figure 1



Oscar François de Jarjayes being presented as a male heir despite being born female.

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This project proposes using two digital humanities tools, the Text Encoding Initiative (TEI) and Voyant Tools, to investigate representations of women in Riyoko Ikeda's Japanese *shōjo* manga, *The Rose of Versailles.* It was serialised in a manga magazine between April 1972 and December 1973 (Anan, 2014, p. 41), a period marked by rapid economic growth in Japan despite the 1973 oil crisis (Tsutsui & Mazzotta, 2014, p. 63). In the 1970s, *Ūman Ribu* (from the English term Women's Liberation) mobilised feminists in Tokyo and many provincial cities, raising new questions about female sexuality, "female nature," and the role of mothers (Chujo & Aizawa, 2020, p. 133).

During the same period, *shōjo* manga, aimed primarily at young female readers, developed significantly and introduced previously unexplored themes (Thorn, 2004). Set during the French Revolution, *The Rose of Versailles* centres on Marie Antoinette and the fictional Oscar François de Jarjayes, a woman raised as a man who serves in the Royal Guards (Anan, 2014, p. 41). The manga follows Oscar's childhood after her father decides to raise her as a boy (Figure 1), her appointment as Marie Antoinette's protector, and her growing disillusionment with the French aristocracy as she becomes increasingly aware of the social inequalities and political tensions that contributed to the French Revolution.

This project asks the following research question: How does *The Rose of Versailles* portray female characters, and what can digital methods reveal about gender roles, relationships, and social expectations in postwar Japan? By combining TEI and Voyant Tools, this project demonstrates both the opportunities and limitations of digital approaches within historical research.

Source Material



Figure 2

Marie Antoinette and Oscar François de Jarjayes, whose relationship forms one of the central narratives of *The Rose of Versailles*.

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This project draws on all chapters of *The Rose of Versailles*, which chart Oscar François de Jarjayes's life from childhood to adulthood. According to Nobuko Anan, this historical manga allowed girls in the 1970s "to experience the life of a revolutionary androgyne who fights to overthrow the ancient, powerful regime of gender and sexuality" (Anan, 2014, p. 41). The series has also been identified as a *yuri* text; Kimberly D. Thompson argues that *yuri* manga "provides fantastical worlds for young readers to explore queer female subjectivities" and classifies *The Rose of Versailles* within this tradition (Thompson, 2017, p. 302). According to Kinko Ito, manga is a highly successful business in Japan, embedded in specific social, cultural, political, and economic contexts and can therefore offer insight into contemporary attitudes toward gender, identity, family, and social roles. It can also depict social phenomena such as racism, classism, and social inequality (Ito, 2005, pp. 456–457).

The project's corpus will consist of transcriptions of dialogue, narration, and other relevant textual elements extracted from all chapters of *The Rose of Versailles*. These transcriptions

will form the basis for textual analysis using TEI and Voyant Tools, enabling the identification of recurring themes, character relationships, and patterns in how the manga represents female identities, particularly in the relationship between Oscar and Marie Antoinette (Figure 2).

What is TEI?

Lou Burnard describes the Text Encoding Initiative (TEI) as one of the most influential projects within the Digital Humanities (Burnard, 2014). According to Burnard, the TEI provides "guidelines for the creation and management in digital form of every type of data created and used by researchers in the Humanities, such as source texts, manuscripts, archival documents, ancient inscriptions, and many others" (Burnard, 2014). TEI enables researchers to encode not only the structure and formatting of texts but also their meaning and content. Furthermore, TEI XML documents are software-independent, allowing researchers to share, preserve, and display them consistently across different computing environments (Burnard, 2014). In this project, TEI will encode dialogue, narration, characters, and relevant themes in *The Rose of Versailles*, creating a structured corpus for subsequent analysis with Voyant Tools.

One major advantage of TEI, Lou Burnard argues, is that it is a community-driven project that evolves in response to users' needs. It also emphasises what is common across all texts, whether digital or non-digital, providing "a kind of encyclopaedia of generally agreed textual notions" (Burnard, 2014). However, TEI has limitations. James Cummings (2019) notes that TEI projects can become large and complex, may be difficult to modify, and require knowledge of XML. Nevertheless, like Burnard, Cummings emphasises TEI's collaborative, community-driven nature, arguing that users can and should make changes and improvements as their needs evolve (Cummings, 2019, pp. 58, 76). TEI uses an encoding language called XML (Extensible Markup Language), a standard developed by the World Wide Web Consortium (W3C). XML represents structured data and lets researchers identify specific textual features using tags. Burnard describes XML as representing data "as a linear stream of character data, and of labelling particular parts of that stream with named tags to indicate structural function or semantics" (Burnard, 2014).

The following example, adapted from Burnard (2014), shows how XML tags can identify structural and semantic elements in *The Rose of Versailles*. In this example, the <persName> tag identifies characters such as Oscar François de Jarjayes and Marie Antoinette, while the <placeName> tag identifies Versailles as a location. The <relationship> tag can also record connections between characters, enabling researchers to analyse patterns of interaction and representation in the text (Burnard, 2014).

```
<?xml version="1.0"?>
<TEI xmlns="http://www.tei-c.org/ns/1.0">
  <text>
    <body>
      <div type="chapter" n="1">
        <sp>
          <speaker>
            <persName gender="female">
              Oscar François de Jarjayes
            </persName>
          </speaker>
          <p>I was raised as a son, yet I remain a woman.</p>
        </sp>

        <p>
          <persName gender="female">
            Marie Antoinette
          </persName>
          speaks with
          <persName gender="female">
            Oscar François de Jarjayes
          </persName>
          about their duties at
          <placeName>Versailles</placeName>.
        </p>

        <relationship type="friendship"
                      active="#Oscar"
                      passive="#MarieAntoinette"/>
      </div>
    </body>
  </text>
</TEI>
```

In this project, similar tags could identify characters, dialogue, locations, relationships, and gender references in *The Rose of Versailles*. By converting the manga's text into a structured dataset, TEI enables researchers to search, organise, and analyse the corpus systematically,

making large volumes of textual material more manageable (cf. Cohen & Rosenzweig, 2006). This approach enables researchers to investigate patterns in female representation, character relationships, and discussions of gender identity that might otherwise be difficult to identify through close reading alone.

What is Voyant?

Voyant Tools is a web-based text analysis tool that generates and visualises statistics on how and how often words are used in a text or corpus. Researchers can upload one or more texts and use a range of built-in tools to examine word frequencies, recurring themes, collocations, and trends across a dataset. By turning textual data into visualisations such as word clouds, frequency charts, and trend graphs, Voyant helps researchers identify patterns that close reading alone may miss (McGraw Center for Teaching and Learning, n.d.).

The TEI-encoded corpus of *The Rose of Versailles* will then be imported into Voyant Tools to analyse patterns in how female characters are represented. For example, Voyant can identify frequently occurring terms associated with Oscar François de Jarjayes and Marie Antoinette, examine how the manga uses gender-related words throughout, and explore relationships between characters through word co-occurrence patterns. Oscar is particularly interesting because, although she is biologically female, she is raised and presented as a man. Analysing the language associated with Oscar may therefore reveal how the narrative constructs gender identity, femininity, masculinity, duty, and social expectations. Comparing the vocabulary used for Oscar with that used for other female characters, such as Rosalie, may also help identify whether the manga portrays her as exceptional or as part of a broader commentary on women's roles in society. These visualisations may help identify recurring themes, character associations, and linguistic trends that are difficult to track consistently through manual reading (McGraw Center for Teaching and Learning, n.d.).

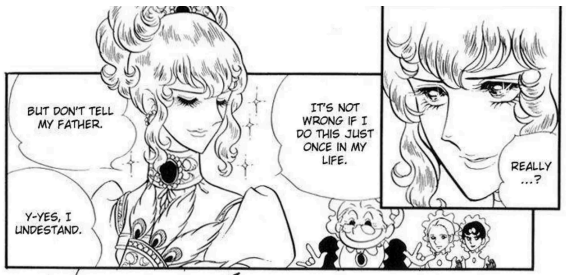


Figure 3

Oscar François de Jarjayes dresses as a woman for one night.

Note. Copyright © Riyoko Ikeda.

One advantage of Voyant Tools is that it lets researchers quickly analyse large amounts of text and present findings through accessible

visualisations. However, it also has limitations. According to the McGraw Center for Teaching and Learning, text-analysis tools are not designed to interpret texts themselves but rather to uncover trends and patterns in language use. Consequently, word frequencies alone cannot explain meaning, context, or authorial intent (McGraw Center for Teaching and Learning, n.d.). For example, they cannot fully capture the significance of Figure 3, in which Oscar dresses as a woman for the first time in her life to attract Count von Fersen, for whom she has developed romantic feelings. Understanding this scene's importance requires attention to character development, visual representation, and narrative context, all of which extend beyond quantitative analysis.

Furthermore, Voyant focuses exclusively on textual data and therefore cannot account for manga's visual elements, such as character design, panel composition, and symbolism, which are often central to interpretation. Oscar's character design is particularly significant because, even when she wears traditionally feminine clothing, she retains many androgynous and "pretty-boy" features that distinguish her throughout the manga (Figure 3). The McGraw Center for Teaching and Learning also notes that text analysis results depend heavily on how a corpus is prepared and segmented; different transcription and encoding choices may therefore influence the findings (McGraw Center for Teaching and Learning, n.d.). As a result, researchers must interpret Voyant's results alongside close reading to avoid reducing complex historical and cultural sources to simple numerical patterns and visualisations.

Methodology

This project will examine all chapters of *The Rose of Versailles* to investigate how the manga represents female characters and constructs gender identities. The analysis will prioritise chapters featuring Oscar François de Jarjayes, Marie Antoinette, and other prominent female characters, such as Rosalie, because these sections contain significant discussions of gender, power, social class, and personal relationships.

The first stage of the project will involve creating a digital corpus of the manga as a whole. Dialogue, narration, and other relevant textual elements will be transcribed from the manga and organised into individual text files. Although *The Rose of Versailles* is a visual medium, this project focuses on its textual content to examine how language expresses gender and identity. Particular attention will be paid to how Oscar presents herself and how others

describe her, including whether she employs traditionally masculine forms of expression or adopts language more commonly associated with female characters.

Once the corpus is created, the texts will be encoded in TEI. Character names will be tagged using <persName>, locations using <placeName>, and significant interactions through appropriate relationship tags. Additional tags will identify references to gender, family roles, authority, and social status where these are relevant to the research question. For example, encoding social status allows us to distinguish between characters such as Oscar, a member of the nobility, and Rosalie, who comes from a much less privileged background. This process will create a structured dataset that can be searched and analysed more efficiently.

After encoding, the TEI files will be imported into Voyant Tools. Several analytical features will then be used to examine patterns within the corpus. The Cirrus word cloud and Terms tools will identify frequently occurring words and concepts associated with major female characters. The Trends tool will examine how the use of specific terms changes across chapters, while the Contexts tool will allow individual words to be studied within their surrounding textual environment. The project will also analyse word co-occurrence patterns to investigate how concepts such as femininity, duty, leadership, authority, and love are associated with particular characters.

The project will pay particular attention to Oscar François de Jarjayes because her position as a woman raised and presented as a man challenges conventional distinctions between masculinity and femininity. Her portrayal combines traditionally masculine attributes, including military service and leadership, with emotional and romantic experiences commonly associated with female characters. By comparing the language used to describe Oscar with that used for other female characters, the project will explore whether the manga presents her as exceptional or as part of a broader commentary on women's social roles. The project will also compare Oscar, Marie Antoinette, and Rosalie to identify differences in how the narrative represents female identity, authority, class, and social expectations. Marie Antoinette is particularly significant because her character develops considerably over the course of the story, transforming from a naïve young woman into a more politically and emotionally complex figure.

The findings from Voyant Tools will then be compared with close readings of selected manga passages. This final stage is essential because quantitative patterns alone cannot explain meaning or historical significance. Instead, the digital analysis results will guide further interpretation and identify themes that warrant closer examination in their narrative context.

Although the narrative is set during the French Revolution, the findings will ultimately be interpreted in relation to the social and cultural context of 1970s Japan, when *The Rose of Versailles* was originally produced and consumed. In doing so, the project explores not only how the story represents female characters but also what those representations may reveal about contemporary discussions of gender, identity, and women's roles in Japanese society.

This methodology combines computational analysis with traditional textual interpretation, allowing the project to investigate both broad patterns and individual examples of female representation in *The Rose of Versailles*.

Why TEI and Voyant Tools?

TEI and Voyant Tools were chosen because they address different aspects of the research process while complementing one another. TEI provides a structured framework for organising textual data and preserving information about characters, locations, social status, and relationships. This transforms the source material into a searchable corpus while retaining important contextual information.

Voyant Tools builds on this foundation by enabling corpus-wide language analysis. While TEI categorises information, Voyant highlights recurring themes, linguistic tendencies, and connections between characters. Together, these tools make it possible to compare individual scenes involving Oscar, Marie Antoinette, and Rosalie with wider trends across the manga.

This combination suits *The Rose of Versailles* because the project examines both individual characters and broader representations of gender. TEI encodes relevant information systematically, while Voyant facilitates exploring how gender, class, authority, and identity are expressed through language. Rather than replacing interpretation, the two tools help direct attention towards themes and relationships that merit further investigation.

Challenges and Limitations

Despite their usefulness, both TEI and Voyant present methodological challenges. Transcription inevitably reduces a visual medium to text. As a result, the corpus may underrepresent important elements of the manga, including panel composition, artistic style, visual symbolism, and character expressions. This matters because visual representation plays a key role in how *The Rose of Versailles* constructs gender and identity.

In addition, the encoding process requires interpretative decisions. Researchers must decide which elements to tag and how to categorise them. Different encoding choices may therefore produce different datasets and influence the analysis results. As Cummings (2019) notes, TEI projects can also become complex and difficult to manage, particularly when large numbers of tags are involved.

Voyant Tools presents further limitations. Although it can identify trends in language use, it cannot independently determine historical significance or interpret meaning. Frequently occurring words are not necessarily the most important, while some significant themes may appear only occasionally. As Milligan (2022) argues, quantitative approaches should therefore complement conventional historical analysis rather than replace it. Likewise, Elo (2020) emphasises that the quality of digital analysis depends heavily on the quality of the underlying data and metadata. Consequently, treat the findings as provisional interpretations rather than definitive conclusions.

Conclusion

This project proposes using TEI and Voyant Tools to investigate representations of women in *The Rose of Versailles*. By combining text encoding and computational analysis, the project aims to identify recurring themes and relationships in the portrayal of female characters, gender identities, and social status.

At the same time, the project recognises the limitations of digital methods. Neither TEI nor Voyant can fully capture the visual complexity of manga, and the interpretation of results ultimately depends on the researcher. Nevertheless, when used critically, digital tools can offer new ways of examining cultural sources and generating historical questions.

By examining *The Rose of Versailles* through both qualitative and quantitative approaches, this project seeks to deepen our understanding of gender representation within the manga and explore how its popularity reflected wider debates about femininity and women's roles in 1970s Japan.

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